

Midterm Exam - November 23rd

Exercise 7. 20 points. Assuming that $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$, prove by the definition with ε - δ that

$$\lim_{t \rightarrow 0} \left(\frac{\sin t}{t} - 1 \right)^2 = 0.$$

Solution: The hypothesis is that $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$, so we use it in the ε - δ - settings.

Let $\varepsilon > 0$, then take $\varepsilon_1 = \sqrt{\varepsilon} > 0$. Due to $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$ for the choosen $\varepsilon_1 = \sqrt{\varepsilon}$ there exists $\delta > 0$ such that

$$0 < |t| < \delta \Rightarrow \left| \frac{\sin t}{t} - 1 \right| < \varepsilon_1.$$

But

$$\left| \frac{\sin t}{t} - 1 \right| < \varepsilon_1 \Rightarrow \left| \left(\frac{\sin t}{t} - 1 \right)^2 \right| = \left| \frac{\sin t}{t} - 1 \right|^2 < \varepsilon_1^2 = \varepsilon.$$

Therefore, for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |t| < \delta \Rightarrow \left| \left(\frac{\sin t}{t} - 1 \right)^2 \right| < \varepsilon.$$

By the **limit definition with** ε - δ the previous sentence means that

$$\lim_{t \rightarrow 0} \left(\frac{\sin t}{t} - 1 \right)^2 = 0. \quad \square$$

Remark: One can take $\varepsilon_1 = \min \{ \varepsilon, 1 \} > 0$ instead of $\varepsilon_1 = \sqrt{\varepsilon} > 0$.

Exercise 8. 20 points. Consider the function for each $\mathbf{c} \in \mathbb{R}$:

$$f(x) = \begin{cases} x^{2/3} & \text{if } x \leq 1 \\ \frac{(x^3 - 5x^2 + 8x - 4\mathbf{c})}{(x-1)} & \text{if } x > 1 \end{cases}$$

a) Is there any $\mathbf{c}$ such that $f'(1)$ exists? Justify your answer using the definition of derivative.

b) Find all $x \in \mathbb{R}$ such that f is differentiable.

Solution. a)

• If f is differentiable at $x = 1$ then f is continuous at $x = 1$, and so, we impose the **necessary** condition of continuity which means

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} f(x) = f(1)$$

Due to the definition of the function we have $f(1) = x^{2/3} \Big|_{x=1} = 1^{2/3} = 1$ then the two side - limits must be equal to 1:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^{2/3} = 1 \quad \checkmark$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{(x^3 - 5x^2 + 8x - 4\mathbf{c})}{(x-1)} \quad ?$$

The last is an indetermination, so the numerator must tend to zero to give us a chance of an existing limit:

$$0 = \lim_{x \rightarrow 1^+} (x^3 - 5x^2 + 8x - 4\mathbf{c}) = 4 - 4\mathbf{c} \Rightarrow \mathbf{c} = 1$$

We compute again the limit from the right with $\mathbf{c} = 1$ and obtain:

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} \frac{(x^3 - 5x^2 + 8x - 4\mathbf{c})}{(x-1)} = \lim_{x \rightarrow 1^+} \frac{(x^3 - 5x^2 + 8x - 4)}{(x-1)} = \lim_{x \rightarrow 1^+} \frac{(x-2)^2(x-1)}{(x-1)} \\ &= \lim_{x \rightarrow 1^+} (x-2)^2 = 1 \quad \checkmark \end{aligned}$$

Hence, only if $\mathbf{c} = 1$ we obtain a **continuous f at $x = 1$** , for which we still

have to verify if f is differentiable or not at $x = 1$, since the continuity is **not enough** to obtain the differentiability.

• If $\mathbf{c} \neq 1$ then f is not continuous at $x = 1$, so f is not differentiable at $x = 1$.

• Consider for $\mathbf{c} = 1$ the **continuous f** defined by:

$$f(x) = \begin{cases} x^{2/3} & \text{if } x \leq 1 \\ (x-2)^2 & \text{if } x > 1 \end{cases}$$

Since f is a piecewise-defined function and the formula changes exactly at $x = 1$, the **differentiability at $x = 1$ can only be verified by definition of the derivative**, that is, we must analyze whether the limit $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$ exists or not; this limit can also be computed in the equivalent form

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{(x - 1)}$$

In order to the limit to exist, the two side-limits should be equal. The **limit of the definition of the derivative** from the left is:

$$\lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{(x - 1)} = \lim_{x \rightarrow 1^-} \frac{x^{2/3} - 1}{(x - 1)} = \lim_{x \rightarrow 1^-} \frac{(x^{1/3} - 1)(x^{1/3} + 1)}{(x - 1)} = \lim_{x \rightarrow 1^-} \frac{x^{1/3} + 1}{x^{2/3} + x^{1/3} + 1} = \frac{2}{3}$$

But this is not equal to the **limit of the definition of the derivative** from the right:

$$\begin{aligned} \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{(x - 1)} &= \lim_{x \rightarrow 1^+} \frac{(x - 2)^2 - 1}{(x - 1)} = \lim_{x \rightarrow 1^+} \frac{x^2 - 4x + 3}{(x - 1)} = \lim_{x \rightarrow 1^+} \frac{(x - 1)(x - 3)}{(x - 1)} \\ &= \lim_{x \rightarrow 1^+} (x - 3) = -2 \end{aligned}$$

The side-limits are different, then **the limit doesn't exist**, so, **$f'(1)$ doesn't exist**, and we conclude that **f is not differentiable at $x = 1$.**

Solution. 8. b) Notice that $f(x) = x^{2/3}$ if $x < 1$ and so

$$\frac{d}{dx} x^{2/3} = \frac{2}{3} x^{-1/3} = \frac{2}{3} \frac{1}{x^{1/3}} \text{ only for } x \neq 0, x < 1$$

In fact, f is not differentiable at $x = 0$, because the corresponding limit of the derivative at $x = 0$ doesn't exist:

$$\nexists f'(0) \text{ because } \nexists \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{x^{2/3} - 0}{x} = \lim_{x \rightarrow 0} \frac{1}{x^{1/3}} = \infty$$

We have proved that **$\nexists f'(0)$ and so, f is not differentiable at $x = 0$.**

Hence, for the derivative we have

$$f'(x) = \begin{cases} \frac{2}{3} x^{-1/3} & \text{if } x < 1, x \neq 0 \\ \nexists f'(x) & \text{if } x = 1 \\ 2(x - 2) & \text{if } x > 1 \end{cases}$$

So, f is neither differentiable at $x = 0$ nor at $x = 1$, hence, the set of x for which f is differentiable is

$$\{x \in \mathbb{R} : x \neq 0, x \neq 1\} = (-\infty, 0) \cup (0, 1) \cup (1, +\infty) \quad \checkmark$$